

Teaching Statement

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Throughout my academic career, I have upheld a teaching philosophy that views education and research as mutually reinforcing pillars of student development. My approach is student-centered, integrating theory with hands-on practice and innovation to build strong foundational knowledge and foster the ability to solve real-world engineering problems independently. This commitment has been recognized through awards such as the Senior Teaching Assistant Award at HKUST, reflecting my dedication to an interest-driven, practice-oriented, and personalized approach to education.

1. Teaching Philosophy

Emphasizing interaction through a student-centred approach. During my tenure as a Teaching Assistant for graduate courses at HKUST, I integrated short quizzes and small-group discussions into tutorials to assess student understanding in real-time and address common difficulties. Furthermore, to cultivate an engineering mindset, I refined course projects by introducing requirements for comprehensive performance evaluation and design justification, encouraging students to balance system-level trade-offs. This initiative was widely adopted by the instructor and significantly improved students' practical competence. In recognition of these contributions, I was honored with the 2023 Senior Teaching Assistant Award.

2. Student Mentorship

Supporting long-term development and career planning. I believe research training extends beyond publications to help students discover their professional path. For instance, I mentored Sida Chen, an undergraduate who was initially enthusiastic but unsure of his research direction. Through close guidance, I helped him align his interest in circuit design with my research needs, leading him to independently implement a board-level power optimization solution for a tactile sensor testing system. This hands-on experience not only solidified his technical expertise but also clarified his commitment to the field. Leveraging the skills gained from this project, he successfully secured admission to the University of Pennsylvania for a master's degree specializing in integrated circuits and power management.

Strengthening process management to maintain project progress. I emphasize timely intervention to prevent junior students from getting stuck in implementation details. For instance, I mentored Jiayi Mao, a visiting first-year PhD student from Westlake University, on a TFT-based neuromorphic circuit project. When she encountered critical discrepancies between experimental results and theoretical models, we established weekly face-to-face meetings to systematically identify root causes and optimize testing strategies. This consistent guidance not only ensured smooth completion of the project but also led to a manuscript currently under review in a high-impact journal, reinforcing the value of regular, deep communication in research training.

3. Leadership and Collaboration

Providing leadership through coordination and role definition. I served as the Vice Chair of the Student Committee for the 10th "Young Leaders in Displays and PG Workshop," co-organizing the event with Chair Runxiao Shi. This workshop attracted over 200 attendees. We coordinated seven oral sessions spanning active-matrix devices and sensors, alongside an

exhibition of over 80 posters. Beyond this, I managed a diverse team of student volunteers to ensure seamless transitions between sessions, honing my skills in project management and cross-disciplinary communication. Additionally, serving as the banquet host allowed me to foster a lively, interactive atmosphere, further strengthening my public speaking and adaptability. The workshop's success deeply reinforced my appreciation for the critical roles of leadership and teamwork in academic organization.

Leveraging strengths for collaborative success. I teamed up with three colleagues for the “ICDT Emphyrean Cup Innovation Competition,” where we secured the Grand Prize for our exceptional work on the full-flow design of a Gate Driver on Array circuit. As a core contributor, I drove a closed-loop design strategy that ensured high consistency between our simulation results and actual device characteristics. This experience not only solidified my mastery of system-level design methodologies spanning from simulation to layout implementation but also honed my ability to balance design trade-offs under strict time constraints. Furthermore, it strengthened my capacity for cross-task coordination and efficient collaboration within a high-pressure competitive environment.

4. Teaching Plans and Course Development

My teaching agenda centers on the fundamental theory, design verification, and fabrication testing of digital and analog circuits, while exploring their seamless integration with MEMS sensor systems. My primary objective is to empower students to translate theoretical knowledge into practical engineering solutions. I emphasize the tradeoffs in circuit design by mastering the balance between area, power, speed, gain, and voltage swing to cultivate strong engineering intuition through iterative experimentation. By bridging the gap between abstract concepts and physical realization, I aim to build a solid foundation for students to become versatile integrated circuit designers capable of tackling complex, interdisciplinary challenges in future engineering projects.

Course 1: Thin-Film Circuit Design and Fabrication

As an advanced extension of conventional analog CMOS curricula, this course focuses on the holistic design, fabrication, and characterization of Thin-Film Transistor (TFT) circuits. Leveraging the university's advanced micro-nanofabrication facilities, the curriculum covers the entire engineering pipeline from device modeling and SPICE simulation to mask layout, physical fabrication, and electrical testing. I emphasize a “concept-to-validation” approach, encouraging students to propose innovative circuit architectures and personally execute the full process from design to measurement. This hands-on training enables students to grasp the critical coupling between manufacturing processes, device structure, and circuit performance, establishing a systemic view beyond discrete components. Assessment will be based on technical results, with outstanding projects recommended for submission to international conferences to bridge classroom learning with broad academic research.

Course 2: Intelligent MEMS Sensors

This course moves beyond the traditional focus on device physics in MEMS microsystems, emphasizing instead the co-design of microsensors, circuits, and systems for intelligent applications. Students will learn foundational theories, analytical design methods, fabrication techniques, and examples of representative devices and products, with the goal of mastering miniaturization technologies. The curriculum covers fundamental mechanics and physics,

micro-fabrication processes, packaging and integration technologies, microsensors, microfluidics, and lab-on-a-chip systems. The course also closely integrates cutting-edge academic research and industry trends, exploring the convergence of MEMS sensors and artificial intelligence in emerging fields such as embodied robotics and biomedical engineering. Through this interdisciplinary approach, the course aims to equip students with the perspective and skills needed to develop deployable smart sensing systems, aligning with advancing trends in the Artificial Intelligence of Things and robotics landscapes.